U/D/TZ

Reg. No: 23BCE2093

Final Assessment Test – April 2026

VIT® Vellore Institute of Technology

Course: BCSE301L - Software Engineering

Class NBR(s): 1368/1369/1391/1397/1401/1409/1413/ 1419/1421/1423/1425/1429/1430/1463/1487/2260/ 4701/6742

Slot: A1+TA1

Time: Three Hours

Max. Marks: 100

- KEEPING MOBILE PHONE/ANY ELECTRONIC GADGETS, EVEN IN 'OFF' POSITION IS TREATED AS EXAM MALPRACTICE
- DON'T WRITE ANYTHING ON THE QUESTION PAPER

COs	CO Statements
CO1	Apply and assess the principles of various process models for the software development.
CO2	Demonstrate various software project management activities that include planning, Estimations, Risk assessment and Configuration Management.
CO3	Perform Requirements modelling and apply appropriate design and testing heuristics to produce quality software systems.
CO4	Demonstrate the complete Software life cycle activities from requirements analysis to maintenance using the modern tools and techniques.
CO5	Escalate the use of various standards and metrics in evaluating the process and product.

BL – Blooms Taxonomy Level (1 – Remember, 2 – Understand, 3 – Apply, 4 – Analyse, 5 – Evaluate, 6 – Create)

Answer ALL Questions
(10 X 10 = 100 Marks)

1. a) A software company develops an online banking system for a bank. After the system is developed, it is installed and used by thousands of customers. Over the years, the system continues to work without any physical wear and tear, but developers regularly update it to fix bugs and add new features like mobile payments and biometric login. The software is very complex. Some parts of the system, such as the login module, are reused in other banking applications. Explain the software characteristics for the above application. [5] CO1 BL2

   b) Describe the evolution of software with an example for each type. [5]

2. Identify the suitable process models for the following applications and justify your answers. CO1 BL3

   i. Payroll management system

   ii. E-commerce websites

   iii. Air traffic control systems

   iv. Social Media Platforms

   v. Highly interactive automobile infotainment system.

3.a) A software company is developing a mobile food delivery application similar to Uber Eats. During the development process, several issues arise like the employees are newly recruited trainees they have no experience in the domain. The client does not explain the system requirements properly and frequently changes features. Team leaders from some teams moved to another company. The project was planned to be completed in schedule, but due to various issues it is delayed. The company needs to spend extra money on additional cloud services from Amazon Web Services to support the application. After the application is launched, many customers continue using established platforms like Uber Eats instead of the new app. Identify any 5 types of risks associated with the given scenario and explain. CO2 BL5

OR

3.b) i. Draw the precedence network diagram for the following data and calculate the completion time. [5] CO2 BL5

ii. Identify the critical path and determine which activitites can be delayed? [5]

Activities	Immediate Predecessor	Duration in weeks		
		To	Tm	Tp
A	-	28	32	36
B	-	22	28	32
C	A	26	36	46
D	B	14	16	18
E	B	32	32	32
F	B	40	52	74
G	D	12	16	24
H	E,G	16	20	26
I	E,G	26	34	42
J	C,F	12	16	30

4.a) The university will require a new Course Registration system. At the beginning of each semester, professors at the university will register the courses that they are going to teach with the system. The system then will allow students to request a course catalogue containing a list of course offerings for the coming semester. Information about each course such as professor, department and prerequisites are included. The student then can select four-course offering. Once the registration process is completed for a student, the registration system sends information to the billing system so the student can pay for the semester. For each semester, there is a period of time that students can change their schedule through adding or dropping courses. Professors must be able to access the system to indicate which courses they will be teaching and to see which students signed up for their course offerings. Based on the above specification. CO3 BL6

i) Create class diagram for the given application with all the relationships mentioned. To illustrate the relationships such as aggregation, association and generalization to specialization using appropriate notations of object oriented Modelling Techniques.

ii) Show the steps involved in course registration and withdrawal process using sequence diagram with all necessary notations.

OR

4.b) Consider a telephone company that has decided to implement an interactive web-based alternative to the telephone directory. The Telephone Directory System (TDS) is to include information about both individual and business customers. Using this service, anybody with access to the Internet shall be able to browse and search the company's list of telephone customers to find their name, address, phone number, and name & type of business (for business customers only). The customers shall be able to query their outstanding balance and get a listing of itemized calls made during a specified period with this system as well. Such queries will be re-directed to an existing accounting software. In addition, customers listed in the directory can extend this information with their email addresses. To access such functionality, customers must authenticate themselves by supplying a password provided by the telephone company. The system must also handle updates of the directory including extension of the customer information with email and web address by company staff. For security reasons, this shall not be done from the web-interface, but only from workstations within the company's internal network. A user friendly interface for the system is crucial for users of arbitrary computer literacy. CO3 BL6

i. Identify the suitable primary and secondary stake holders of the system. [2]

ii. List minimum of four functional requirements against each stakeholder. Suggest any four non-functional requirements should be incorporated to improve the quality of the system. [8]

5. Consider a bank ATM as an application example and elaborate any five UID (User Interface Design) principles applied in the design of the application. CO4 BL3

6. Identify and elucidate the types of coupling and cohesion. CO4 BL3

i) An InputHandler module has readFromFile(), readFromDB(), and readFromAPI() — all selected based on a passed-in type flag.

ii) A CustomerOnboarder module does getCustomer(), applyDiscount(customer) and sendWelcome(customer) — all acting on the same customer data.

iii) An OrderProcessor module reaches into the internals of an InventoryManager module and directly alters its private stock counter variable, bypassing any method.

iv) calculateTax(price: number, taxRate: number): number — only the exact required values are passed and returned.

v) OrderService publishes an 'OrderPlaced' event to a message queue. ShippingService subscribes and reacts. Neither knows about the other's internals or even that the other exists.

7. a) Construct the control flow graph and calculate the cyclomatic complexity. [5] CO5 BL5

b) Identify the independent paths and justify how basis path testing helps to ensure that the program is thoroughly tested with minimum number of test cases. [5]

```
def sample_function(x):

if x > 0:
y = x + 1
else:
y = x - 1
if y > 0:
z = y * 2
else:
z = y / 2
return z
```

8. Apply a test plan for the scenario given in Q. No.4b. Identify the key features to be tested and elaborate the testing strategy required for Telephone Directory System. CO5 BL5

9. A company developed an Online Film Ticket Booking System that follows legacy system that allows users to select movies, select the seat and payment methods. The system did not undergo any modification in the recent years. It should be tailored to the need of the time. Suggest the possible upgrades the system may require to meet the new user requirements. Analyse the changes in accordance with the given example. Identify and Justify all the types of software maintenance that are applicable. CO4 BL3

10. a) Discuss about any two of six sigma quality for software management to eliminate defects in the products and processes. [5] CO4 BL2

b) Compare and Contrast CMM, CMMI with respect to the Key process Areas. [5]

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